

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
RUMOMASI**

TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: SEVEN

CLASS: SS2

SUBJECT: Chemistry

TOPIC: Metals & Their Properties

Introduction

- Metals are elements that usually lose electrons to form positive ions. About 80% of the elements are metals; they are found mainly in the earth's crust (ores) and in seawater.

Physical properties

- They are lustrous (shiny), malleable and ductile; good conductors of heat and electricity; sonorous; have high melting and boiling points (except mercury, a liquid); and are dense. They crystallise in metallic lattices with a 'sea' of delocalised electrons, which explains their conductivity and malleability.

Chemical properties

- They react with oxygen to form basic oxides ($4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$); with water (sodium and potassium react violently; calcium slowly; iron rusts in moist air); with dilute acids to give hydrogen ($\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$); and with chlorine to give salts. The reactivity series: K, Na, Ca, Mg, Al, Zn, Fe, Sn, Pb, H, Cu, Ag, Au — a more reactive metal displaces a less reactive one from its salt solution.

Alloys

- An alloy is a mixture of a metal with other metals or a non-metal, made to improve properties: brass (copper + zinc), bronze (copper + tin), stainless steel (iron + chromium + nickel), duralumin (aluminium + copper + magnesium).

CLASS EVALUATION

- State four physical properties of metals.
- Arrange in order of decreasing reactivity: Fe, Zn, Na, Cu, Al.
- What happens when sodium is dropped in water?
- Why are alloys made? Give two examples.

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: EIGHT

CLASS: SS2

SUBJECT: Chemistry

TOPIC: Extraction of Metals

Introduction

- Most metals occur as compounds (ores) because they are reactive; only the noble ones (gold, silver, copper) occur native. Extraction is the process of obtaining the pure metal from its ore.

Principles

- 1. Concentration of the ore (remove gangue by washing, froth flotation or magnetic separation). 2. Reduction to the metal — by carbon (the blast furnace for iron; the cheapest method for moderately reactive metals), or by electrolysis (for the very reactive metals — aluminium, sodium). 3. Refining to high purity (electrolysis, e.g. for copper).

The blast furnace (iron)

- The furnace is charged with iron ore (haematite Fe_2O_3), coke and limestone; hot air is blasted at the bottom. Coke burns to CO_2 , then CO (the reducing agent): $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$. Limestone removes the sandy impurities as slag (calcium silicate); molten iron runs out at the bottom and is cast into pig iron, which is converted to steel in the basic oxygen furnace.

Aluminium by electrolysis

- Bauxite (Al_2O_3) is purified, then dissolved in molten cryolite (to lower the melting point) and electrolysed: aluminium forms at the cathode, oxygen at the anode. It is expensive because of the huge electricity demand — hence aluminium smelters are sited near cheap power.

CLASS EVALUATION

1. Define an ore.
2. State the three steps of metal extraction.
3. Write the equation for the reduction of haematite.
4. Why is electrolysis used for aluminium?

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: NINE

CLASS: SS2

SUBJECT: Chemistry

TOPIC: Carbon & Its Compounds

Introduction

- Carbon (Group 4) is the basis of organic chemistry. It exists as different allotropes — diamond, graphite and the fullerenes — and forms millions of compounds.

Allotropes

- Diamond** — every carbon bonded to four others in a giant tetrahedral lattice: the hardest natural substance, not a conductor, used for cutting tools and jewellery.
- Graphite** — layers of hexagonal rings with weak forces between them: soft, slippery, conducts electricity (delocalised electrons), used in pencils, lubricants and electrodes.
- Fullerenes** — hollow cage molecules (buckminsterfullerene C₆₀).

Oxides of carbon

- Carbon dioxide (CO₂)** — formed by burning carbon, respiration and fermentation: turns limewater milky, used in fire extinguishers and fizzy drinks; it is a greenhouse gas.
- Carbon monoxide (CO)** — formed by the incomplete combustion of carbon: poisonous (combines with haemoglobin), formed in car exhausts and from charcoal stoves in unventilated rooms.

Carbonates and hydrocarbons

- Carbonates (CaCO₃, Na₂CO₃) react with acids to give CO₂; test: add dilute HCl, bubble the gas into limewater. Hydrocarbons (alkanes, alkenes) — the fuels and feedstocks of the petrochemical industry.

CLASS EVALUATION

- Differentiate between diamond and graphite.
- How do you test for CO₂?
- Why is carbon monoxide dangerous?
- Give two uses of graphite.

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: TEN

CLASS: SS2

SUBJECT: Chemistry

TOPIC: Nitrogen & Its Compounds

Introduction

- Nitrogen makes up about 78% of the air. Its compounds — ammonia, nitric acid and the nitrates — are vital for life (proteins) and for agriculture (fertilisers).

Nitrogen and ammonia

- Nitrogen is a colourless, odourless, unreactive gas — so unreactive that it is used to provide an inert atmosphere (food packaging, lamps). It is obtained from air by fractional distillation and is used to make ammonia by the Haber process: $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$ (450 °C, 200 atm, iron catalyst). Ammonia is a pungent gas, very soluble in water, forming the weak alkali ammonium hydroxide — the basis of ammonium fertilisers.

Nitric acid and nitrates

- Nitric acid (HNO_3) is made in industry by the Ostwald process (ammonia → nitric oxide → nitrogen dioxide → HNO_3). It is a strong acid, an oxidising agent, and reacts with metals and carbonates. Nitrates are very soluble salts — ammonium nitrate, urea and NPK are the fertilisers that feed world farming; the nitrate ion is tested by adding sodium hydroxide and aluminium (ammonia is evolved) or by the brown-ring test.

The nitrogen cycle

- Nitrogen is fixed from the air by lightning and by nitrogen-fixing bacteria (free-living and in legume root nodules); plants take up nitrates; animals eat plants; decomposers return ammonia; nitrifying bacteria convert ammonia to nitrite and nitrate; denitrifying bacteria return nitrogen to the air.

CLASS EVALUATION

1. State the percentage of nitrogen in air.
2. Write the Haber process equation with conditions.
3. Why are nitrates important in agriculture?
4. Sketch the nitrogen cycle briefly.

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: ELEVEN

CLASS: SS2

SUBJECT: Chemistry

TOPIC: Sulphur & Its Compounds

Introduction

- Sulphur is a yellow solid non-metal found as the free element (in volcanic regions — Sicily, Japan) and in sulphide ores (chalcocite, zinc blende) and sulphates (gypsum). It exists as rhombic and monoclinic allotropes and is obtained by the Frasch process (superheated water melts it underground and it is forced to the surface).

Uses and compounds

- Sulphur is used in gunpowder, matches, fungicides, rubber vulcanisation and the manufacture of sulphuric acid. It burns with a blue flame to give sulphur dioxide (SO_2) — a pungent gas that bleaches and kills bacteria. SO_2 dissolves to give the weak acid sulphurous acid; with oxygen and a catalyst (V_2O_5) it gives SO_3 , which dissolves in water to give sulphuric acid (the Contact process).

Sulphuric acid

- H_2SO_4 is the most important industrial chemical: fertilisers (superphosphate), car batteries (the accumulator), detergents, paints, dyes and the pickling of metals. Its properties: a dense, oily, corrosive liquid; a strong acid; a dehydrating agent (charring sugar); and an oxidising agent with hot concentrated acid.

Sulphate tests

- Sulphate ions give a white precipitate of barium sulphate with barium chloride solution (insoluble in dilute HCl). Sulphites and hydrogen sulphide are also tested — H_2S has a rotten-egg smell and gives a black precipitate of lead sulphide with lead nitrate.

CLASS EVALUATION

1. State the Frasch process briefly.
2. What are two uses of sulphur?
3. Describe the Contact process.
4. How is a sulphate ion tested in the laboratory?

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: TWELVE

CLASS: SS2

SUBJECT: Chemistry

TOPIC: Chemistry of the Environment

Introduction

- This week applies the chemistry of the term to the environment: the air, pollution and the greenhouse effect — and how chemical knowledge helps protect the planet.

Air and the greenhouse effect

- Clean dry air: 78% nitrogen, 21% oxygen, about 1% argon, 0.04% carbon dioxide, with traces of other gases, water vapour and dust. CO₂ and water vapour (and methane) are greenhouse gases: they trap outgoing heat, keeping the earth warm. The burning of fossil fuels has raised CO₂ levels — the enhanced greenhouse effect warming the planet, melting glaciers and raising sea levels.

Pollutants and their chemistry

- **Carbon monoxide (CO)** — incomplete combustion; a poison. **Oxides of nitrogen (NO_x)** — from vehicle engines; form acid rain and photochemical smog. **Sulphur dioxide (SO₂)** — from burning coal and oil; acid rain which corrodes buildings and kills fish and forests. **Chlorofluorocarbons (CFCs)** — deplete the ozone layer (old refrigerants and aerosols). **Smoke, soot and particulates** — respiratory problems.

Control measures

- Catalytic converters in cars (converting CO and NO_x to CO₂ and N₂); catalytic scrubbers in factories (lime absorbs SO₂); clean fuels and renewable energy (hydro, solar, wind); public transport and energy conservation; reforestation; and international agreements (the Kyoto and Paris agreements).

CLASS EVALUATION

1. Name the main gases of clean air.
2. What are the major greenhouse gases?
3. Explain how acid rain is formed and two of its effects.
4. State three ways of reducing air pollution.